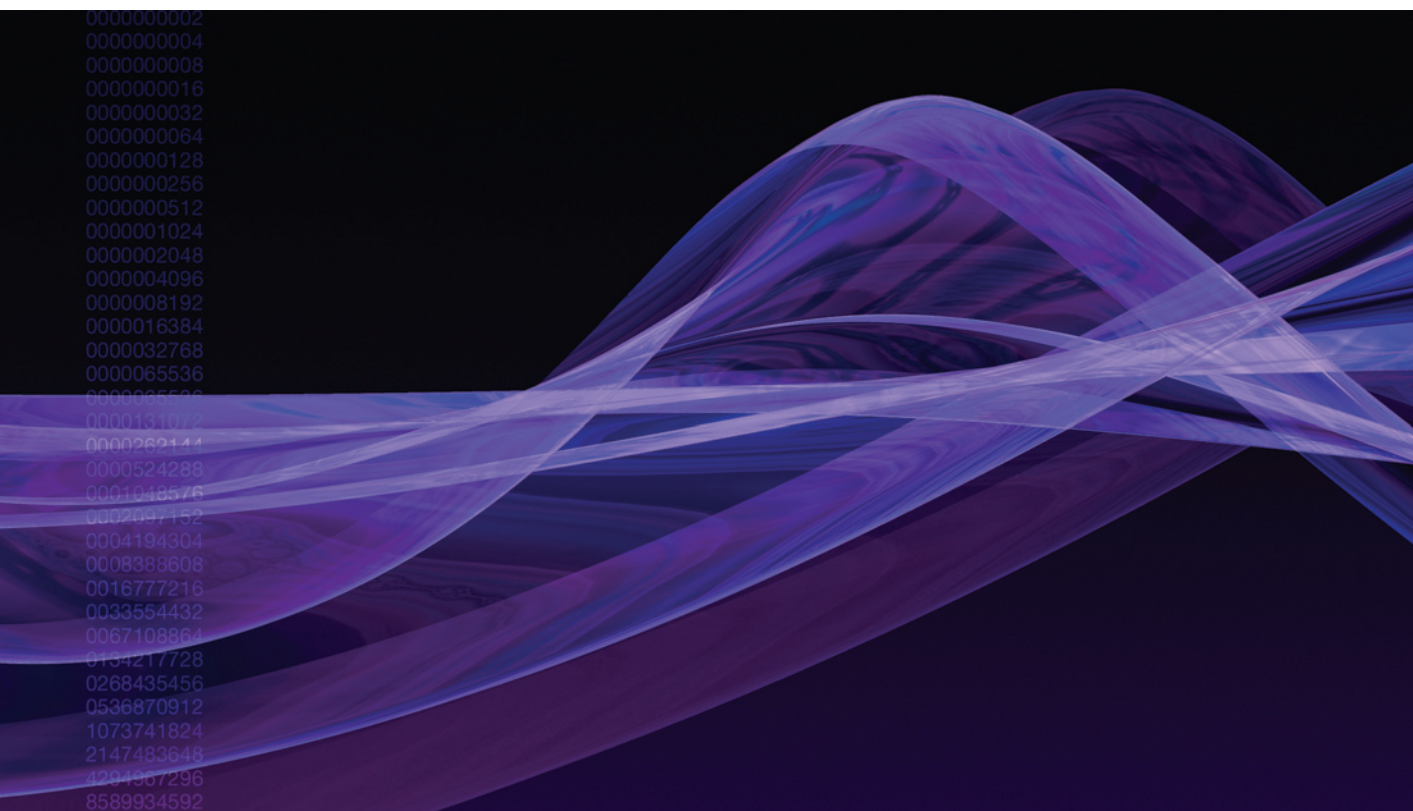


Amplification: Chromo4 Real-Time PCR System



Chromo4™
Four-Color Real-Time PCR System

BIO-RAD



Building on a Flexible Platform

The DNA Engine® family of thermal cyclers from Bio-Rad offers precise temperature control and a modular design that can serve as a platform for future advances. The Chromo4 detector is one such advance, continuing the tradition of innovation by bringing real-time quantitative PCR to the DNA Engine and Dyad Disciple™ platforms.



Flexibility and Thermal Precision

The Chromo4 detector is a real-time detector mated to a 96-well Alpha™ unit. This design allows the Chromo4 detector to be swapped with any Alpha unit on any DNA Engine or Dyad Disciple chassis, with no tools required. The flexibility of the system is maintained, so differently formatted Alpha units can be swapped back in for other applications.



Because the Chromo4 detector snaps into the thermal cycler, this real-time PCR system has the same high performance, precise thermal control, and gradient capability for which the DNA Engine family is renowned. The accurate, uniform thermal control yields reliable, reproducible results, and the temperature gradient feature allows you to optimize cycling conditions in a single experiment.



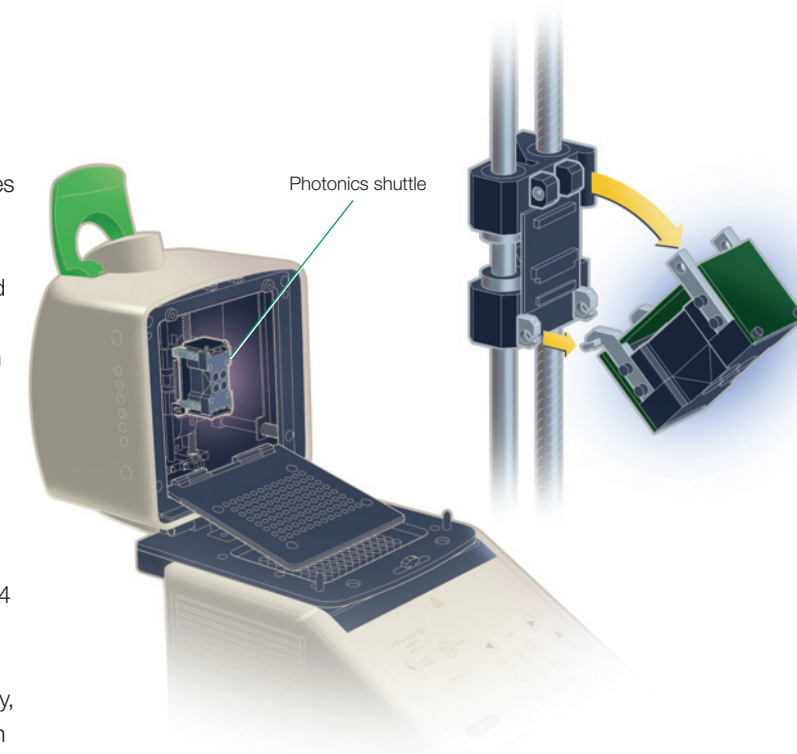
Multiple options. Run one Chromo4 detector on a DNA Engine chassis or two Chromo4 detectors on a Dyad Disciple chassis. With two Dyad Disciple cyclers, up to four Chromo4 detectors can be controlled by a single computer.

Modular Photonics

Modularity is incorporated into the design of the Chromo4 detector itself. The Chromo4 photonics shuttle, which houses all the optical components for real-time detection, quickly snaps in and out of the detector. The standard filter set supports most commercially available dyes, but if you need specialized filter sets for unique chemistries, you can order customized photonics shuttles. The interchangeable design of the photonics shuttle makes such changes simple.

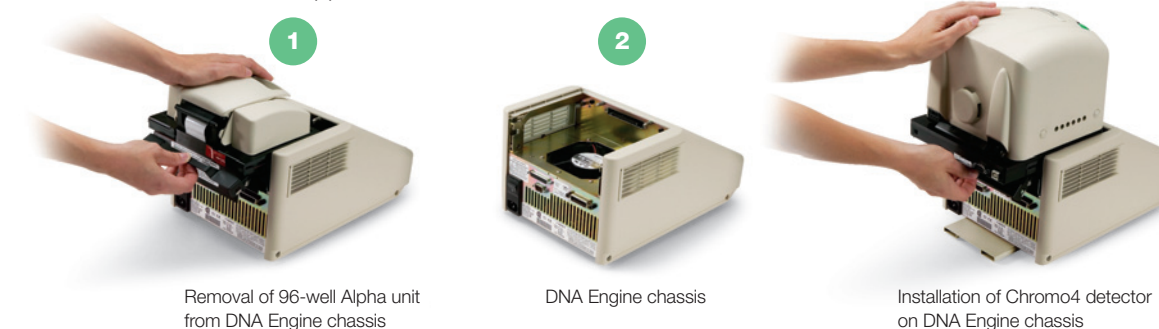
A Platform for Future Advances

Like the DNA Engine family of cyclers, the modular design of the Chromo4 system allows it to be adapted to new technologies and applications without major instrument redesign. It will be possible, for example, to use the Chromo4 detector with the DNA Engine Dyad® and DNA Engine Tetrad® 2 cyclers. When new photonics shuttles become available, you'll be able to upgrade your system economically, without having to buy an entirely new instrument. And with each addition, you will continue to receive the performance features you expect from the DNA Engine family.



Interchangeable formats for changing demands

The interchangeable-module format enables the DNA Engine cycler to keep pace with rapidly changing demands in the laboratory. The Chromo4 detector can quickly and easily be swapped with an Alpha unit to accommodate different applications.



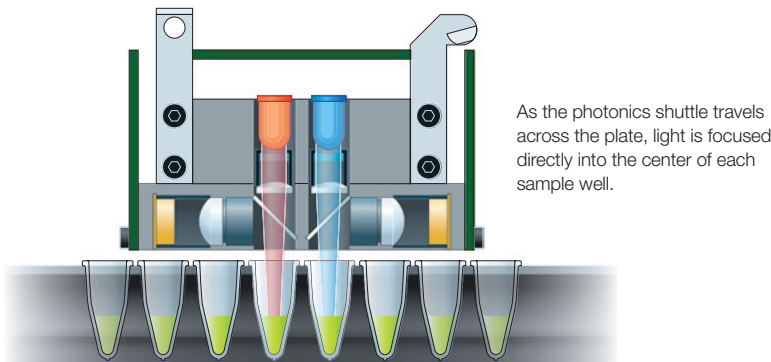
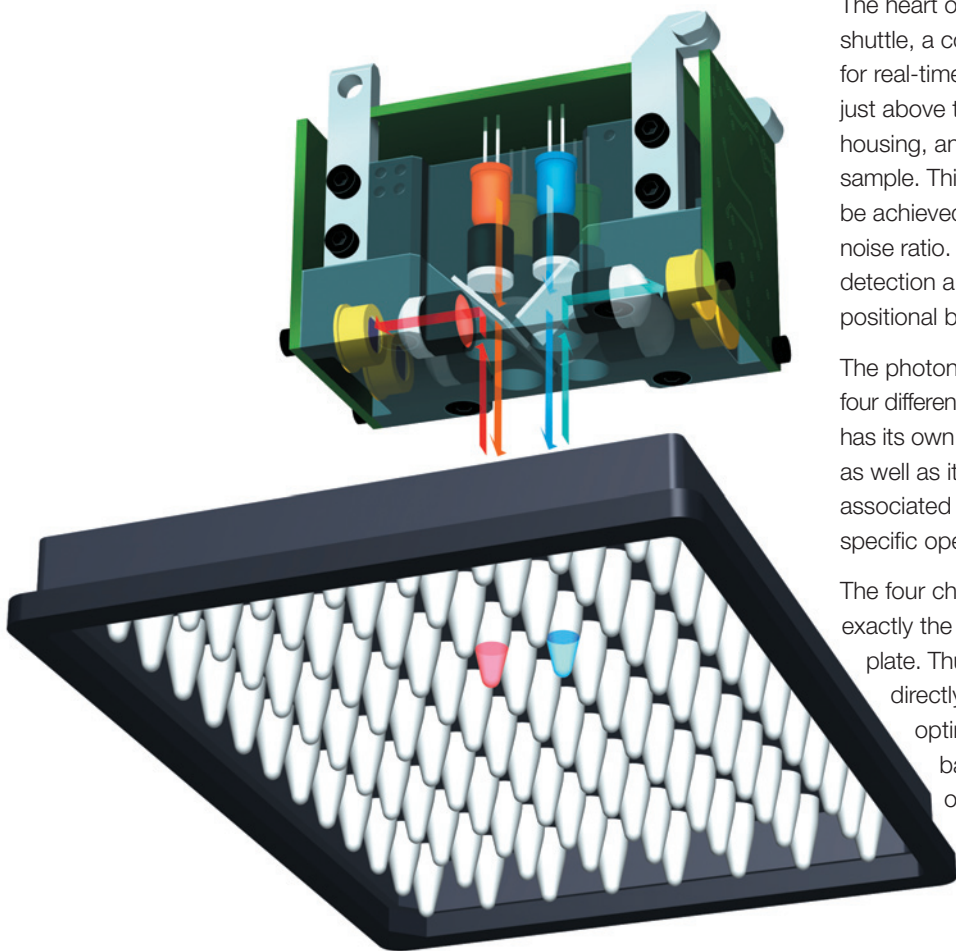
Compact, Sensitive Photonics

The Photonics Shuttle

The heart of the Chromo4 detector is the photonics shuttle, a compact unit that contains all the components for real-time excitation and detection. The shuttle rides just above the sample plate, on axes inside the Chromo4 housing, and it individually illuminates and detects each sample. This unique design allows maximum sensitivity to be achieved with minimal cross talk, for a high signal-to-noise ratio. And because the light paths for excitation and detection are exactly the same for each well, there is no positional bias to worry about.

The photonics shuttle has four channels to detect up to four different colors in the same sample well. Each channel has its own filtered light-emitting diode (LED) for excitation, as well as its own filtered photodiode for detection. Light associated with each channel passes through a channel-specific opening in the base of the photonics shuttle.

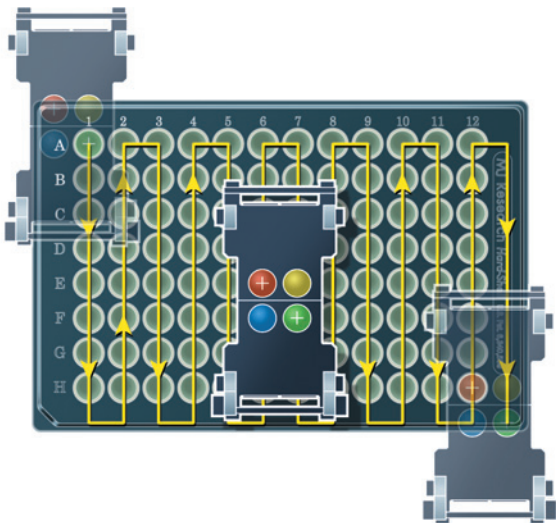
The four channel openings in the photonics shuttle have exactly the same spacing as the wells of the sample plate. Thus, the excitation light from each LED passes directly down into the center of each sample well for optimal excitation. Emitted light passes directly back into the shuttle to the photodiodes for optimal sensitivity.



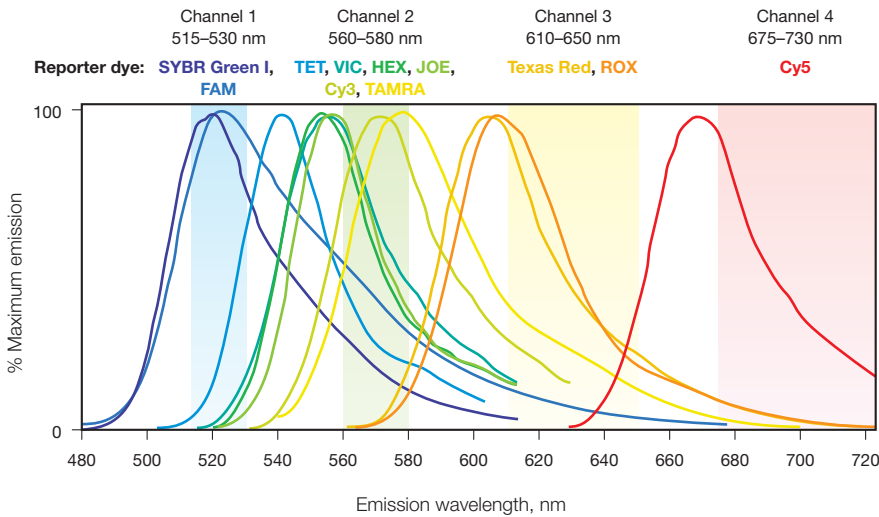
During a plate read, the photonics shuttle travels in a serpentine path across the sample plate, pausing above each well to excite and detect each sample. At every well, the four LEDs fire sequentially, each exciting a different sample. Then the shuttle moves to the next position and repeats the process, reading all 96 wells in all four colors in about 10 seconds. At every position and with every scan, the shuttle is reproducibly centered above each set of wells, so the light path is always optimal.

Less Downtime, More Experiments

The Chromo4 system uses solid-state components for both excitation and detection: LEDs and photodiodes, respectively. Because solid-state components are small and durable, the Chromo4 system can pack all its power into a compact, reliable instrument that requires little maintenance.

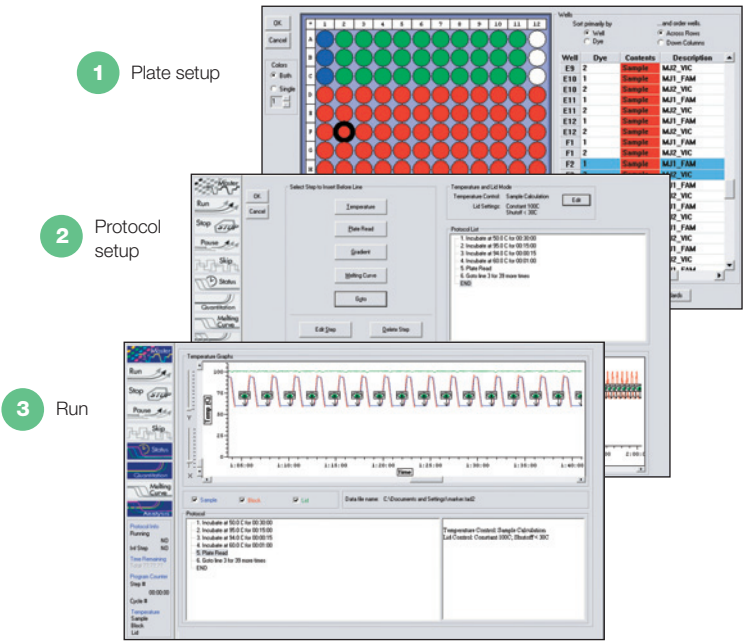


The serpentine path of the photonics shuttle. The shuttle scans an entire plate, independently exciting and detecting each well, in just seconds.

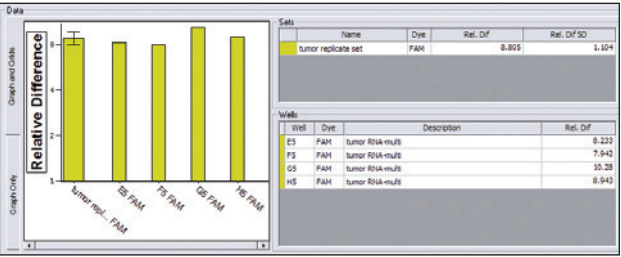


The standard photonics shuttle has four filters that allow detection of most commercially available dyes. The system is precalibrated for use with SYBR Green I, FAM, TET, VIC, HEX, JOE, Cy3, TAMRA, Texas Red, ROX, and Cy5 dyes. The standard set should be sufficient for most researchers' needs, but customized filter sets can be designed for use with unique chemistries.

Researcher-Oriented Software and Reproducible Results



With Opticon Monitor software, you can go from start to run in three easy steps.



Relative gene expression analysis. Graph shows the expression ratio of ERBB2 in tumor vs. normal tissue, calculated using the $\Delta\Delta C_T$ method. Four replicates are shown, along with their average (leftmost bar). Error bars on leftmost bar represent the standard deviation. A FAM-labeled hydrolysis probe was used to detect ERBB2 amplification, and a VIC-labeled probe was used to detect amplification of the housekeeping gene GAPDH, which was used for normalization.

The Chromo4 system is controlled by Opticon Monitor™ software, which makes data collection and analysis easy. Go from start to run in just three steps, observe data acquisition in real time, and begin data analysis while the run is in progress.

Easy Setup

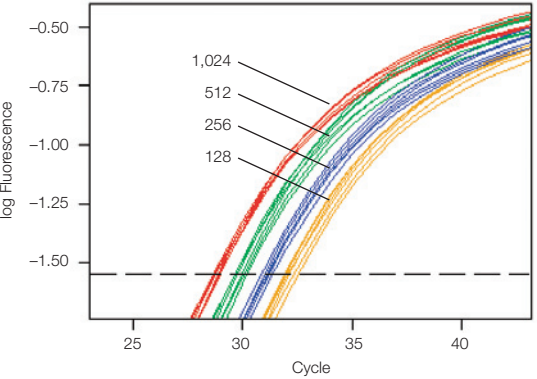
Experimental setup can be done quickly with **Plate** and **Protocol** setup screens. You can easily select the wells you plan to use in the plate diagram, and identify their contents by clicking on buttons for sample, standard, and blank. Protocol creation is just as simple: Select temperature incubation, plate read, thermal gradient, melt-curve, and cycling steps by clicking on the corresponding button, and you will be prompted to enter the appropriate parameters. Once you click **Run**, a status window appears so you can track the progress of the experiment and stop it as soon as sufficient data have been collected.

Easy Analysis

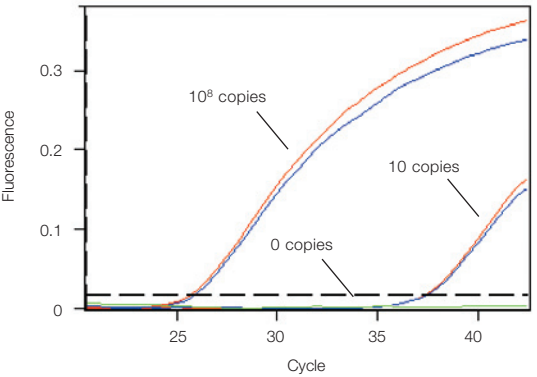
As soon as you begin collecting data, you can begin to analyze it — using a variety of methods. Opticon Monitor software offers a choice of methods for subtracting background signals and for positioning the threshold line. Once these are chosen, the software automatically generates a standard curve and quantitates the template in samples of unknown concentration. The software can also perform relative gene expression analysis using the $\Delta\Delta C_T$ method. Finally, Opticon Monitor software makes it easy to incorporate a melt-curve step to verify product identity.

Clear Results

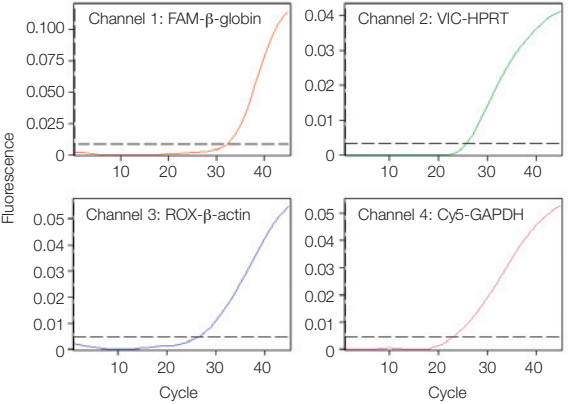
The Chromo4 system generates high-resolution data in four colors. Because the scanning photonics shuttle brings the components for excitation and detection directly to each well, the system is maximally sensitive and robust. Measurements are extremely uniform across all wells, and small differences in copy number can be resolved. Whatever your real-time application — gene expression analysis, genotyping, pathogen detection, or another — the compact, easy-to-use Chromo4 system delivers the performance you need.



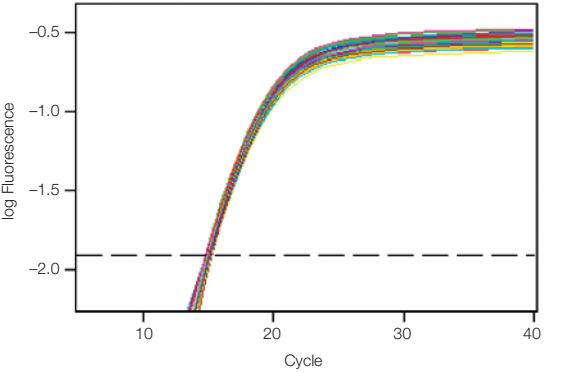
High resolution. Plots of log fluorescence vs. cycle number for amplification of 2-fold serial dilutions of plasmid containing human β -actin cDNA. Product was detected using a FAM-labeled hydrolysis probe. Initial template copy number is indicated for each dilution set. Experiments were performed in replicate ($n = 6$). Note the clear separation of ~ 1 C_T value between dilution sets.



Low cross talk. Fluorescence intensity traces from qPCR in which a high-copy (10^8 copies) sample and low-copy (10 copies) sample were run in adjacent (—) or nonadjacent (—) wells. The C_T values were nearly identical regardless of position, indicating that there was no significant cross talk between the adjacent wells. No-template control (—) showed no significant fluorescence. Template was β -actin plasmid; the indicating fluorophore was FAM.



Four-color multiplexing. Fluorescence intensity traces from each of four channels are shown for a fourplex experiment. The reaction contained four different plasmids and primer pairs, as well as four sequence-specific hydrolysis probes. The probes detected in each channel are indicated on graphs. The initial copy number for each plasmid type was as follows: β -globin, 10^3 ; HPRT, 10^3 ; β -actin, 10^4 ; GAPDH, 10^6 .



Uniformity. Plots of log fluorescence from 95 replicate amplification reactions show the consistency in C_T values across wells. (Single no-template control reaction not shown.) Plasmid containing human β -actin was amplified and detected using a SYBR Green I qPCR kit. Initial template number = 10^6 . C_T values varied $<1.0\%$ across 95 samples.

Specifications and Ordering Information

Chromo4 System

Sample capacity	96-well microplate or 12 x 0.2 ml 8-tube strips
Sample volume	10–100 µl (20 µl recommended)
Dimensions (W x D x H)	20 x 23 x 18 cm (excluding thermal cycler)
Weight	6.8 kg (excluding thermal cycler)
Fluorescence excitation range	Channel 1: 450–490 nm; channel 2: 500–535 nm; channel 3: 555–585 nm; channel 4: 620–650 nm
Fluorescence detection range	Channel 1: 515–530 nm; channel 2: 560–580 nm; channel 3: 610–650 nm; channel 4: 675–730 nm
Linear dynamic range of starting copy number	Up to 10 orders of magnitude
Detection limit of template starting copy number	Down to single-copy detection

DNA Engine and Dyad Disciple Thermal Cyclers

Operational

Number of cycling bays	DNA Engine: 1 Dyad Disciple: 2
Input power	DNA Engine: 100–240 VAC, 50–60 Hz, 850 W max. Dyad Disciple: 200–240 VAC, 50–60 Hz, 1,600 W max., fitted with NEMA L6-20P plug
Speed of ramping	Up to 3°C/sec
Temperature range	0–105°C
Temperature accuracy	Average temperature within ±0.3°C of programmed value at 90°C, NIST-traceable
Temperature uniformity	±0.4°C within 30 sec of arrival at 90°C

Temperature Gradient

Gradient accuracy	±0.3°C of programmed target at end columns, within 30 sec after gradient step timer starts, NIST-traceable
Column uniformity	±0.4°C, well-to-well within column, within 30 sec of reaching target temperature
Gradient range	30–105°C
Temperature differential range	1–24°C
Gradient calculator accuracy	±0.4°C of actual well temperature

Minimum Computer Requirements

Operating system	Windows XP Professional
Memory	512 MB RAM
Processor speed	1.5 GHz
Storage	20 GB hard drive
Screen resolution	1,024 x 768 pixels
Graphics	Graphics card with >64 MB RAM

Catalog # Description

Chromo4 System and Components*

CFB-3260G	Chromo4 Four-Color Real-Time PCR System , includes optical housing, photonics shuttle, DNA Engine® thermal cycler, 96-well sample block, analysis software
CFB-3240G	Chromo4 Four-Color Real-Time PCR Detector , includes optical housing, photonics shuttle, 96-well sample block, analysis software (complete system requires PTC-0200G or PTC-0221G)
PTC-0221G	Dyad Disciple Dual-Bay Thermal Cycler Chassis , does not include Alpha units (requires 2)
CFO-3202	Desktop Computer , includes 2.4 GHz processor, 512 MB RAM, 40 GB hard drive, 48 x 24 x 48 CD-RW, Windows XP Professional operating system
OCM-3201	15" Flat-Screen Computer Monitor
CFO-3203	Laptop Computer , includes 2.4 GHz processor, 512 MB RAM, 40 GB hard drive, 24 x 10 x 24 CD-RW, Windows XP Professional operating system

*The Chromo4 detector can be used with existing DNA Engine and Dyad Disciple cyclers with a free software upgrade.

Purchase of this instrument conveys a limited non-transferable immunity from suit for the purchaser's own internal research and development and for use in applied fields other than Human In Vitro Diagnostics under one or more of U.S. Patents Nos. 5,656,493, 5,333,675, 5,475,610 (claims 1, 44, 158, 160–163 and 167 only), and 6,703,236 (claims 1–7 only), or corresponding claims in their non-U.S. counterparts, owned by Applied Biosystems. No right is conveyed expressly, by implication or by estoppel under any other patent claim, such as claims to apparatus, reagents, kits, or methods such as 5' nuclease methods. Further information on purchasing licenses may be obtained by contacting the Director of Licensing, Applied Biosystems, 850 Lincoln Centre Drive, Foster City, California 94404, USA.

Bio-Rad's Chromo4 real-time thermal cyclers are licensed real-time thermal cyclers under Applied's United States Patent No. 6,814,934 B1 for use in research and for all other fields except the fields of human diagnostics and veterinary diagnostics.

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